

Statistics Superpowers

and other leftovers

👨‍🏫 Statistical Reasoning Lecture #6
Alexander Savi, 2025
📚 None

Untitled by Katharina Brunner (generativeart package) 🎨

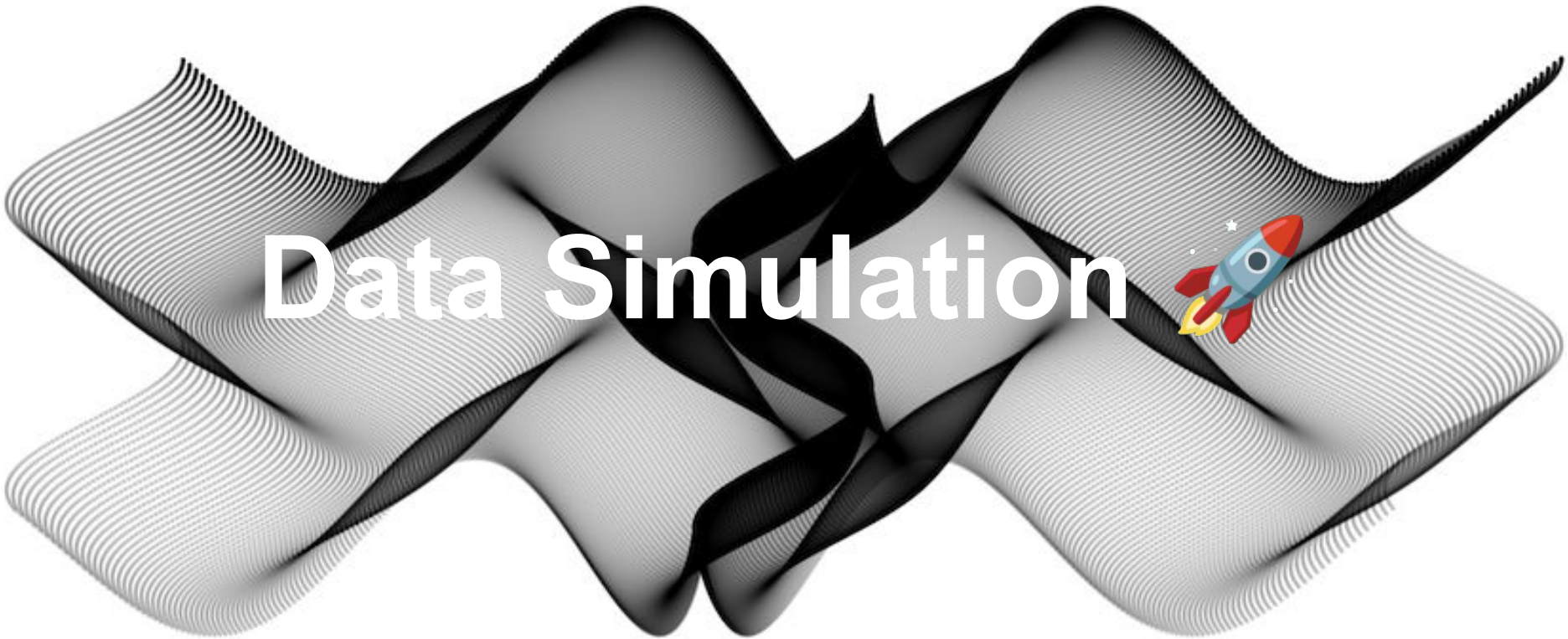
📢 Announcements

- ❏ No R & Tidyverse questions this week
- ❏ New Test Choice questions this week
- ❏ New quiz format that doesn’t leak answers

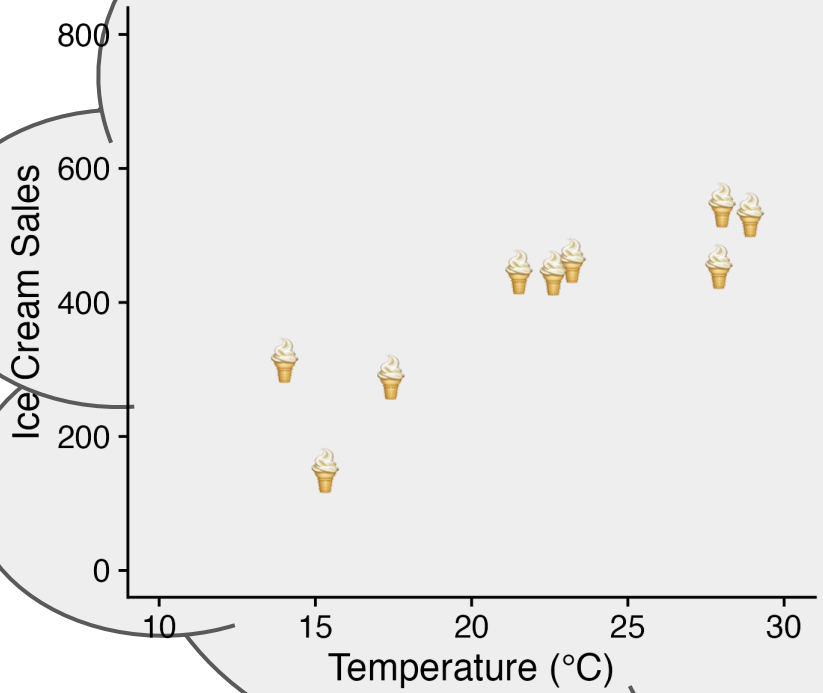
🔮 Today

Topics

1 | Statistical reasoning with GLM
2 | Multiple linear regression
3 | Dummy-variable regression
4 | Bayesian statistics
5 | Logistic regression
6 | Multilevel and longitudinal analysis
7 | Statistics superpowers
    - 7.1 | Data simulation
    - 7.2 | A priori power analysis
    - 7.3 | Polynomial regression
    - 7.4 | Nesting (revisited)
    - 7.5 | Assumptions


Data Simulation

Data ~~collection~~simulation




```
ice_cream_data <- tibble::tribble(
  ~temperature, ~ice_cream_sales,
  27.9,         452,
  15.3,         148,
  17.4,         287,
  21.5,         444,
  28.2,         935,
  14,           312,
  28,           544,
  28.9,         529,
  23.2,         461,
  22.6,         442)
```

$$\text{ice_cream_sales} = \beta_0 + \beta_1(\text{temperature}) + \epsilon$$

```
mod <- ice_cream_sales ~ temperature  # y ~ x
```

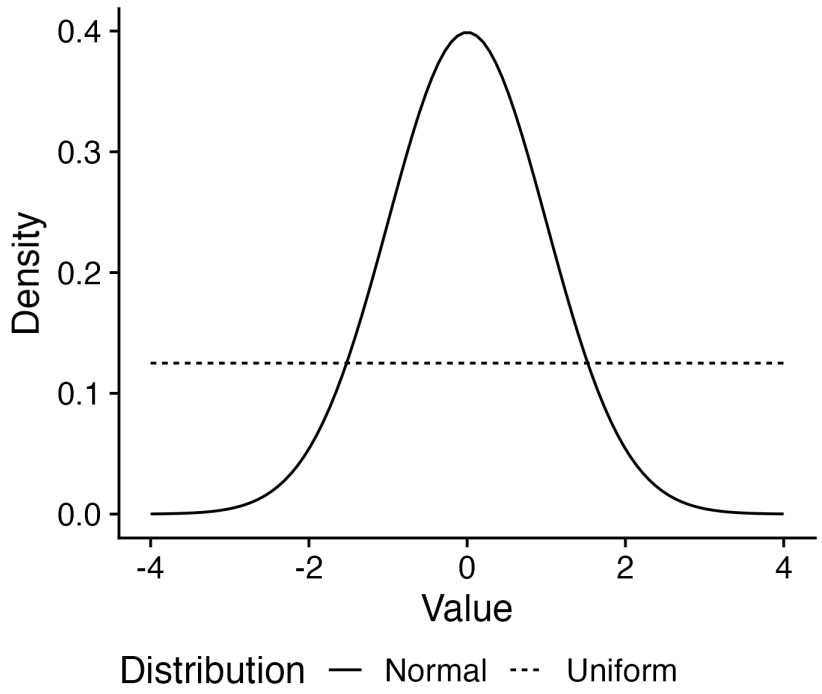
Simulation #1 | Ice cream data

```
set.seed(0)

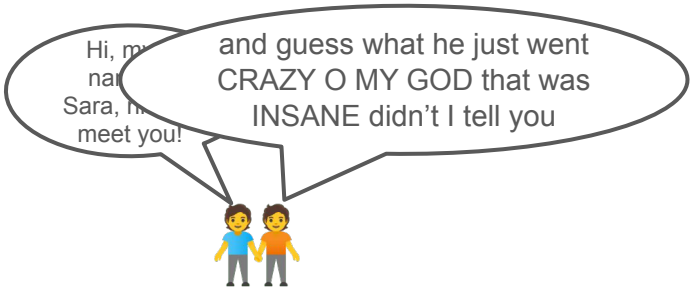
n <- 10
temperature <- runif(n = n, min = 10, max
= 30)

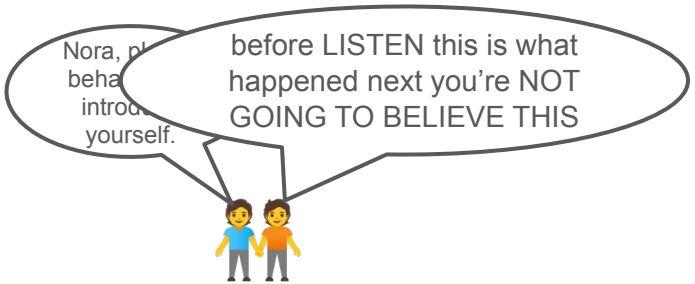
intercept <- -200
slope <- 30
noise_sd <- 120
ice_cream_sales <- intercept + slope *
temperature + rnorm(n = n, mean = 0, sd =
noise_sd)

data <- tibble(temperature,
ice_cream_sales)
```



Signal and noise



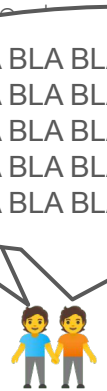


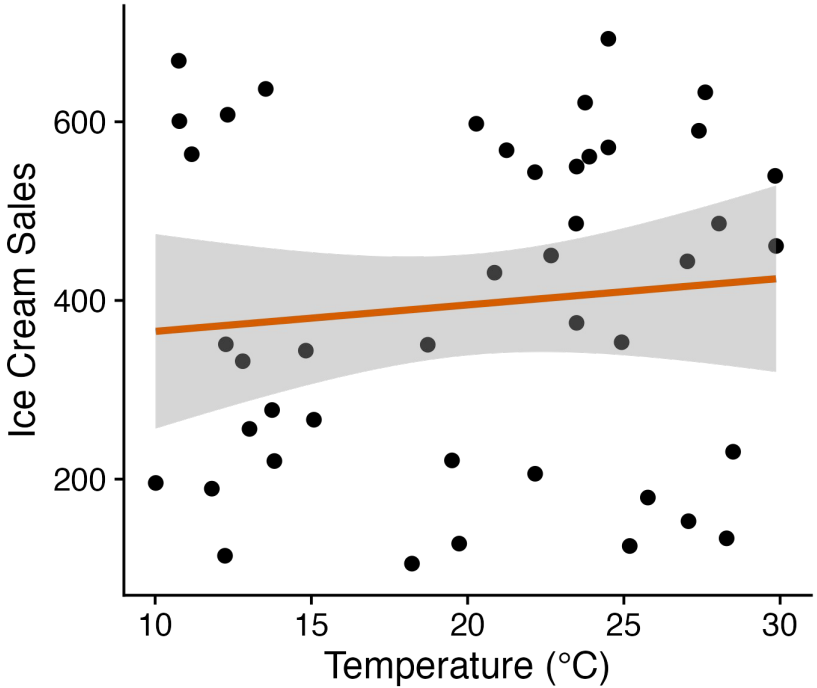
Simulation #2 | High noise, low noise

```
noise_sd_high <- 500
ice_cream_sales_sd_high <- intercept +
slope * temperature + rnorm(n = n, mean =
0, sd = noise_sd_high)

noise_sd_low <- 5
ice_cream_sales_sd_low <- intercept +
slope * temperature + rnorm(n = n, mean =
0, sd = noi
```

BLA BLA BLA BLA BLA BLA
BLA BLA BLA BLA BLA BLA BLA BLA
BLA BLA BLA BLA BLA BLA BLA BLA
BLA BLA BLA BLA BLA BLA BLA BLA
BLA BLA BLA BLA BLA BLA





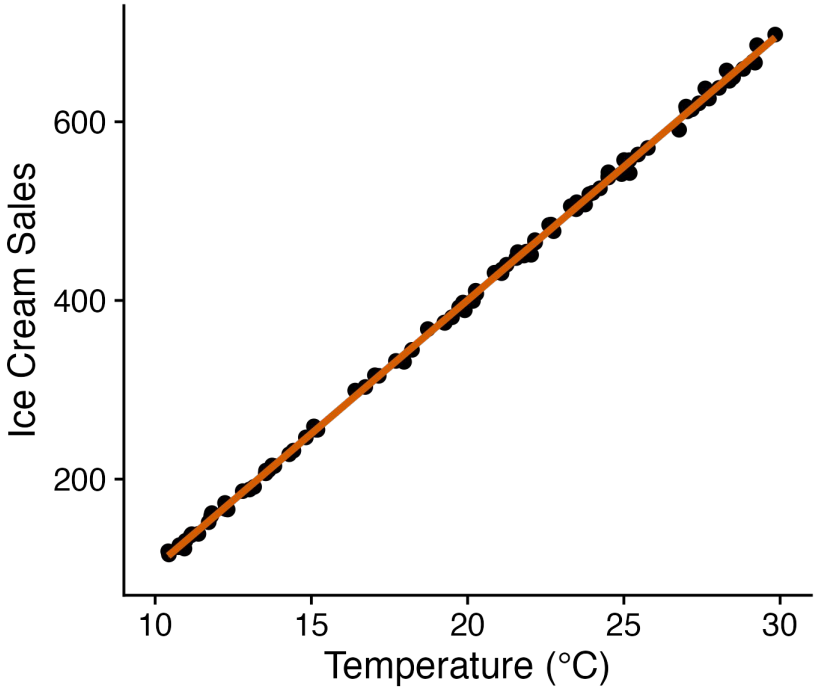
🎵 Master Peace & Georgia - I Might Be Fake

Simulation #2 | High noise, low noise

```
noise_sd_high <- 500
ice_cream_sales_sd_high <- intercept +
slope * temperature + rnorm(n = n, mean =
0, sd = noise_sd_high)

noise_sd_low <- 5
ice_cream_sales_sd_low <- intercept +
slope * temperature + rnorm(n = n, mean =
0, sd = noise_sd_low)
```





🎵 Master Peace & Georgia - I Might Be Fake

Simulation #3 | Frequentist inference (NHST)

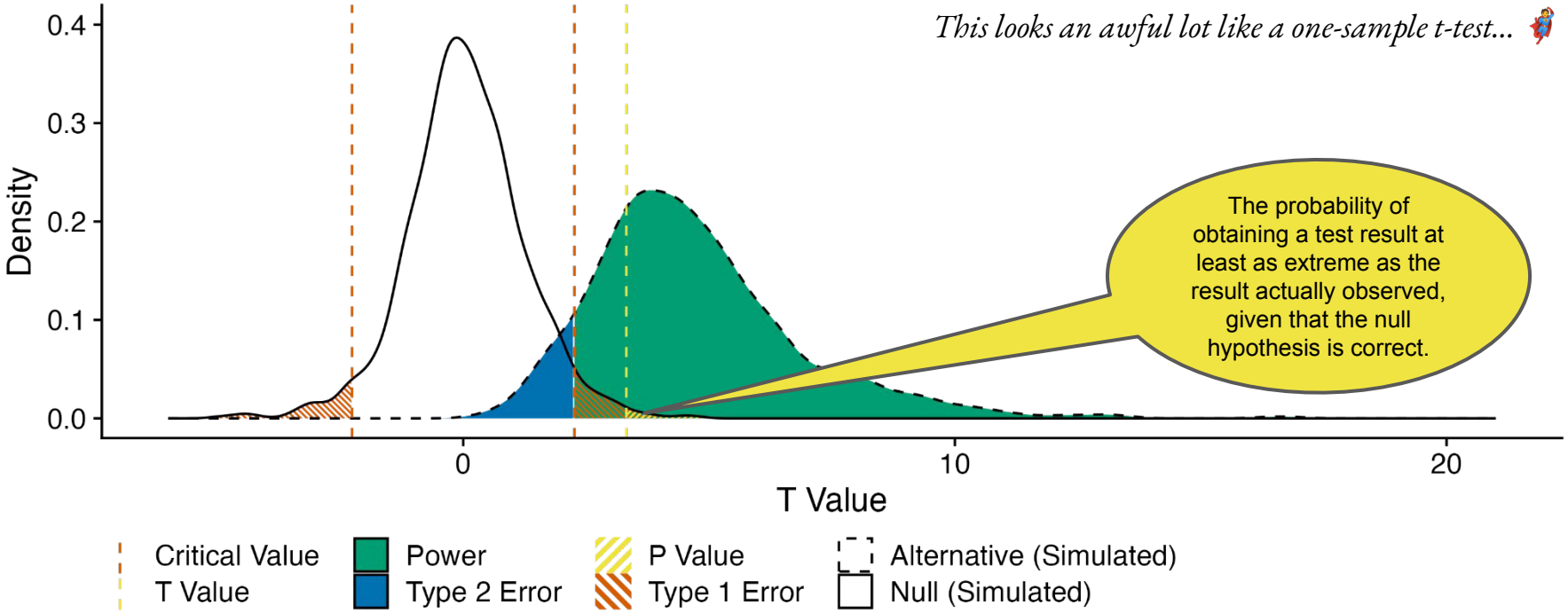
```
n_sim <- 1000
slope_null <- 0
slope_alt <- slope

null_t_stats <- numeric(n_sim)
for (i in 1:n_sim) {
  obs_temperature <- runif(n = n, min = 10, max = 30)
  ice_cream_sales <- intercept + slope_null * obs_temperature
  noise <- rnorm(n = n, mean = 0, sd = noise_sd)

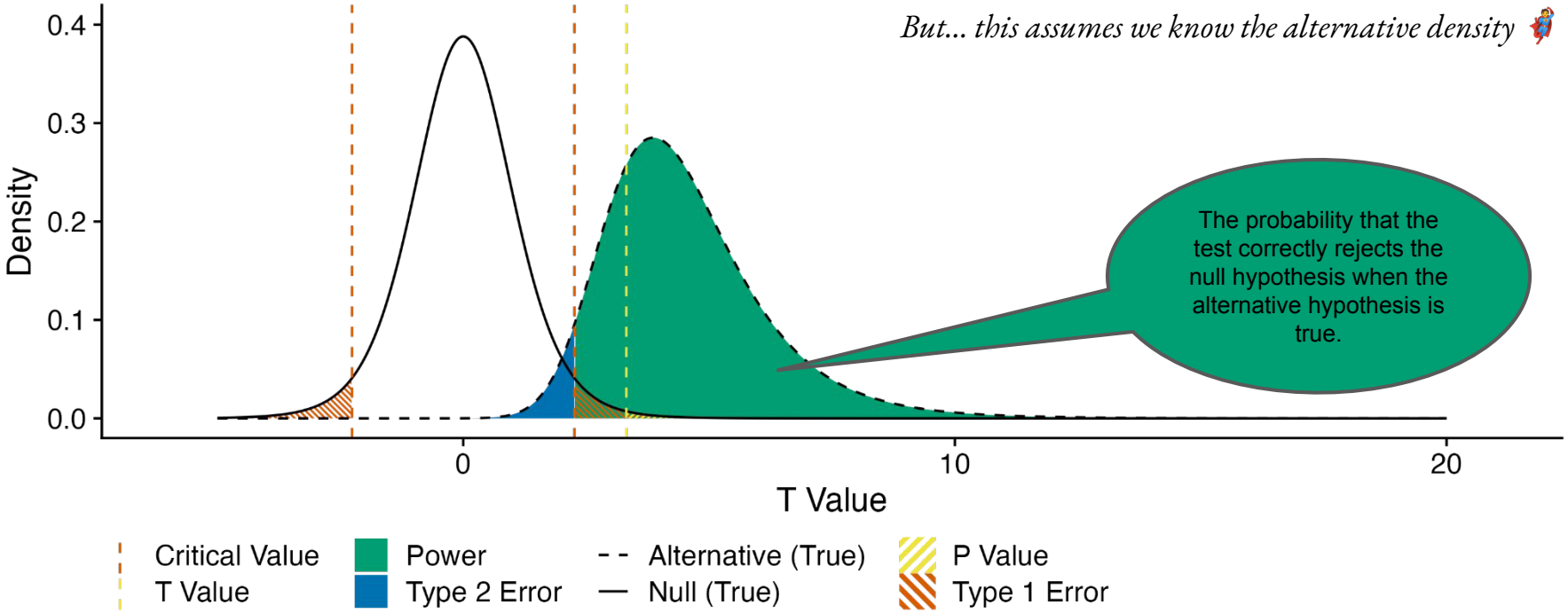
  data <- tibble(
    temperature = obs_temperature,
    sales = ice_cream_sales + noise)

  fit <- lm(sales ~ temperature, data = data)
  null_t_stats[i] <- summary(fit)$coefficients[2, "t value"]
}  # and do the exact same for the alternative slope
```

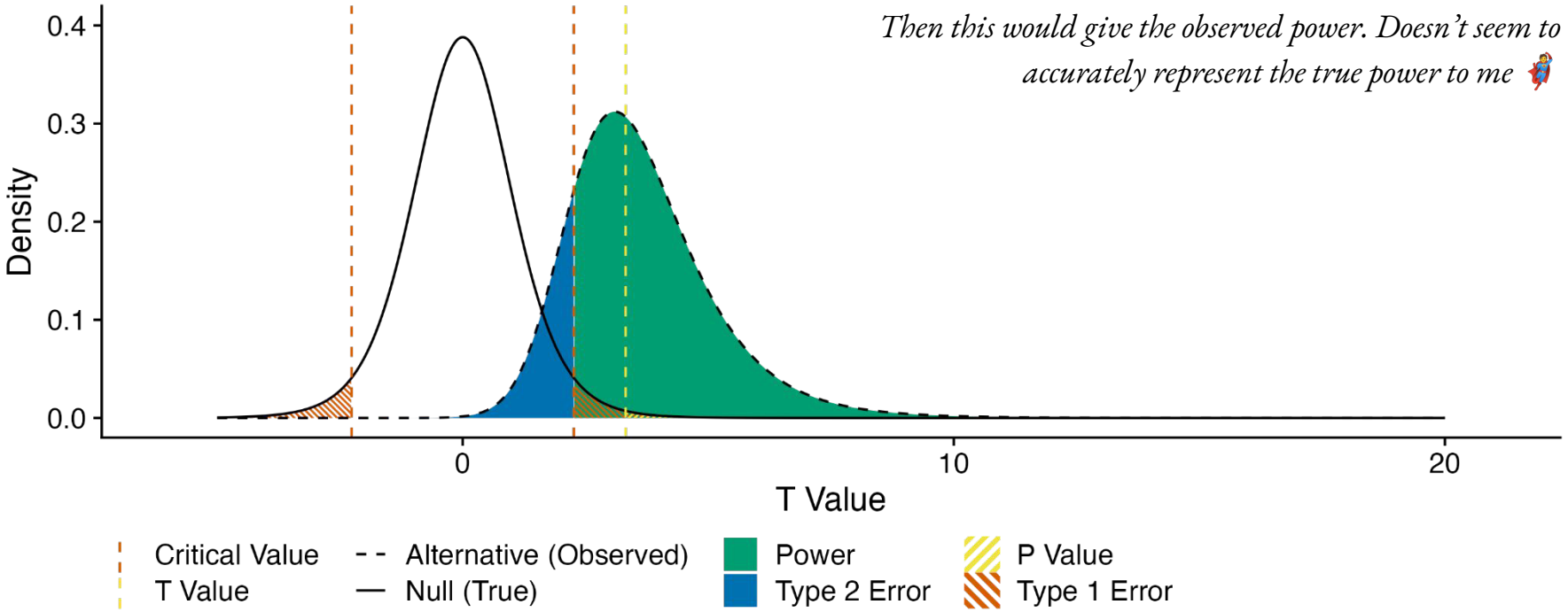
Simulated densities



True densities

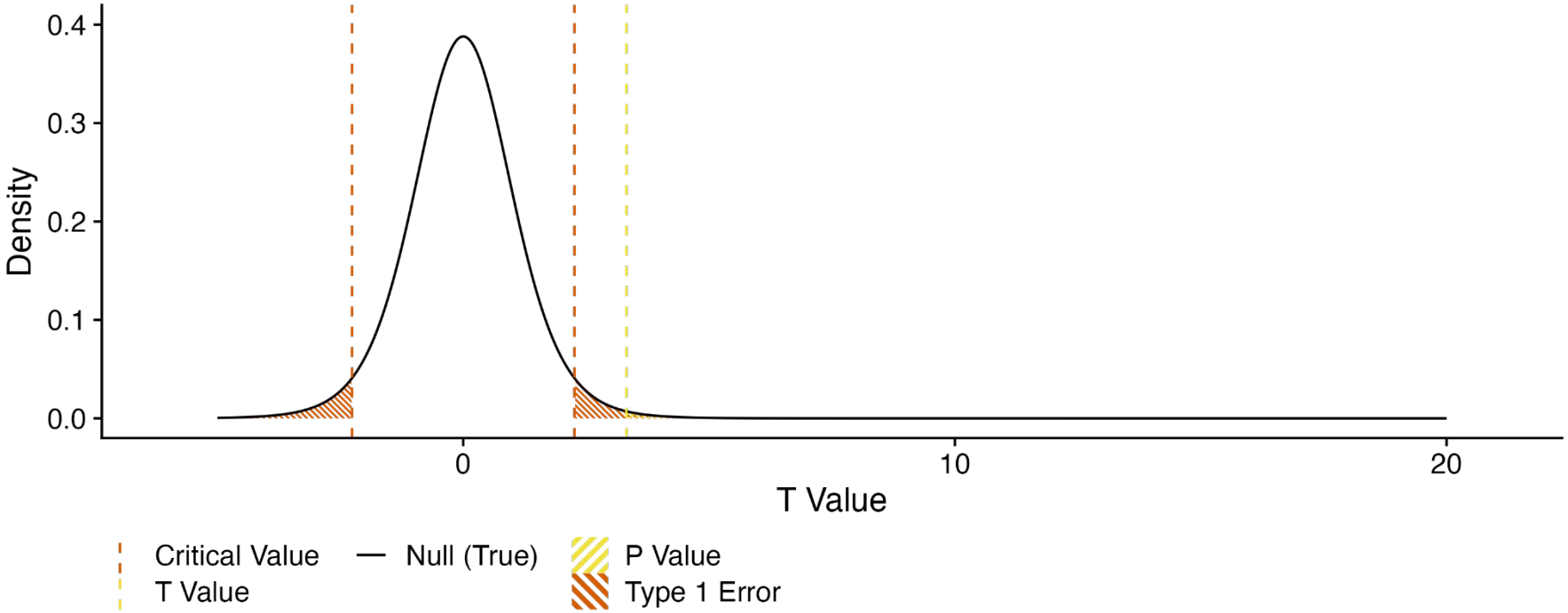


Observed densities



Then this would give the observed power. Doesn't seem to accurately represent the true power to me

NHST in practice



Simulation #3 | Confidence intervals (CI)

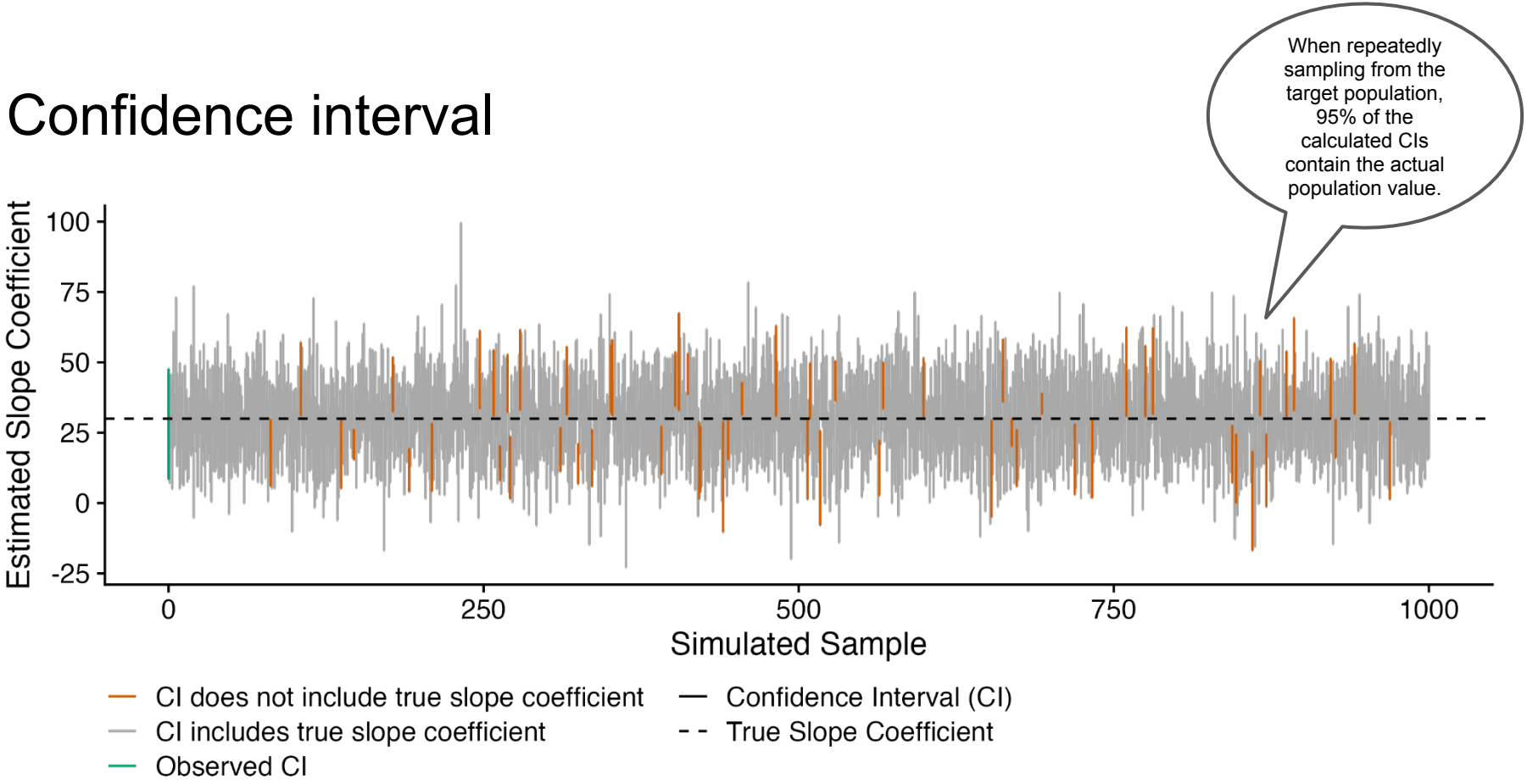
```
n_sim <- 1000
slope_alt <- slope

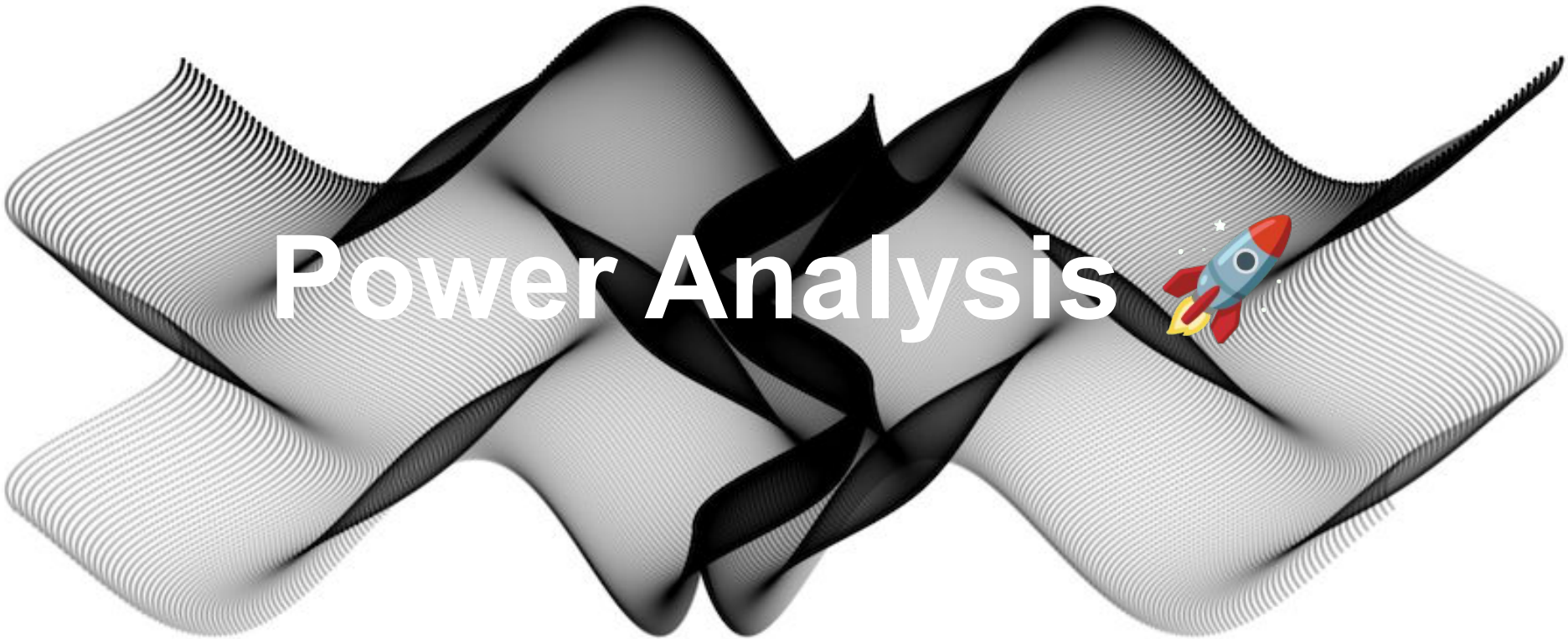
alt_ci <- matrix(NA, nrow = n_sim, ncol = 2)
for (i in 1:n_sim) {
  obs_temperature <- runif(n = n, min = 10, max = 30)
  ice_cream_sales <- intercept + slope_alt * obs_temperature
  noise <- rnorm(n = n, mean = 0, sd = noise_sd)

  data <- tibble(
    temperature = obs_temperature,
    sales = ice_cream_sales + noise)

  fit <- lm(sales ~ temperature, data = data)
  alt_ci[i, ] <- confint(fit)[2, ]
}
```

Confidence interval




Power Analysis

Manipulating statistical power

🦹 *Let's increase* ***α***

And accept more false positives? 🦸

🦹 *Let's perform* ***a one-tailed test***

I don't have a hypothesis about the effect's direction 🦸

🦹 *Let's increase* ***the magnitude of the effect***

Interesting, but we're analyzing observational data 🦸

🦹 *Let's decrease* ***the noise***

Nora: NOOOOOOOO! I HAVE FEELINGS TOO! 👫
Shut up, Nora, I'll try to use preciser measurements 🦸

🦹 *Let's use a* ***more powerful test***

I'm already using the most powerful test 🦸

🦹 *Let's increase* ***the sample size***

Data collection is costly and takes time, how many observations do I need? 🦸

Iris







Q. Are the dimensions of the petals and sepals of the iris flower related?

H. The length of a petal is related to the length and the width of a sepal.

E. […]

💡 A data set made famous by Ronald Fisher and with its very own Wikipedia page.

A priori power | How many observations do I need?

What effect size do I want to be able to detect?

- ❏ Pick values from multiple robust studies
- ❏ Pick magnitude of practical interest
- ❏ Do not pick from a single noisy study

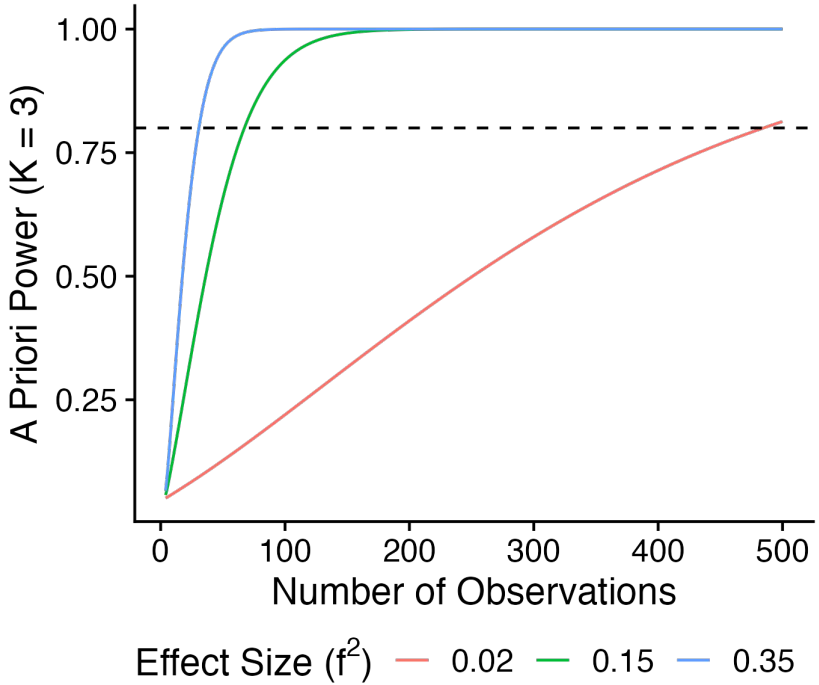
What power do I want to detect an effect?

- ❏ e.g., ≥ .8 (a type II error is 4 times as acceptable as a type I error: .2 vs. .05)

What's the required significance level?

- ❏ e.g., .05

What's the required sample size?



📖 not in book; Gelman, Hill, & Vehtari (2021, p. 293)

A priori power | How many observations do I need?

What effect size do I want to be able to detect?

- ❏ .15 (rule of thumb for medium effect)

What power do I want to detect an effect?

- ❏ .8

What's the required significance level?

- ❏ .05

What's the required sample size?

- ❏ 67.319

```
library("pwr")
a_priori_power <- pwr.f2.test(
  u = 2,  # determined by the model of choice
  v = NULL,
  f2 = .15,
  sig.level = 0.05,
  power = 0.8)

required_n <- ceiling(a_priori_power$v +
a_priori_power$u + 1)
```

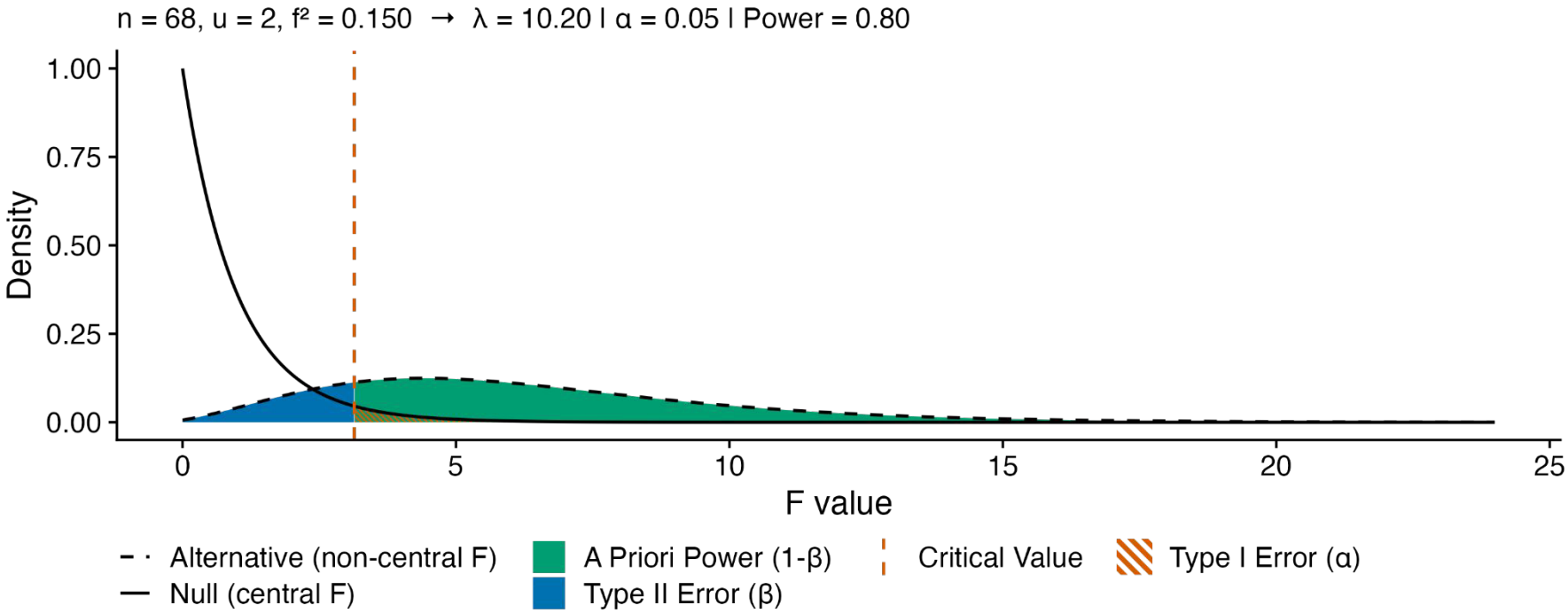
where `u` equals the numerator df of the F test:

- ❏ *1* for each continuous predictor
- ❏ *n levels − 1* for each categorical predictor
- ❏ for each interaction, the product of the df of the terms involved

📖 not in book; Gelman, Hill, & Vehtari (2021, p. 293)

⚠️ for ordinary least squares with independent observations

F distribution



Minimal detectable effect size | What can I detect?

What sample size do I have?

- 150

What power do I want to detect an effect?

- .8

What's the required significance level?

- .05

What effect size am I able to detect?

- .07 (in between small and medium effect)

```
library("pwr")
mde <- pwr.f2.test(
  u = 2,
  v = 150 - 2 - 1,
  f2 = NULL,
  sig.level = 0.05,
  power = 0.8)

mde$f2
```

where `v` equals the denominator df of the F test:

- $n - u - 1$ when testing the whole model
- n is number of observations
- u is the numerator df from above
- the 1 accounts for the intercept

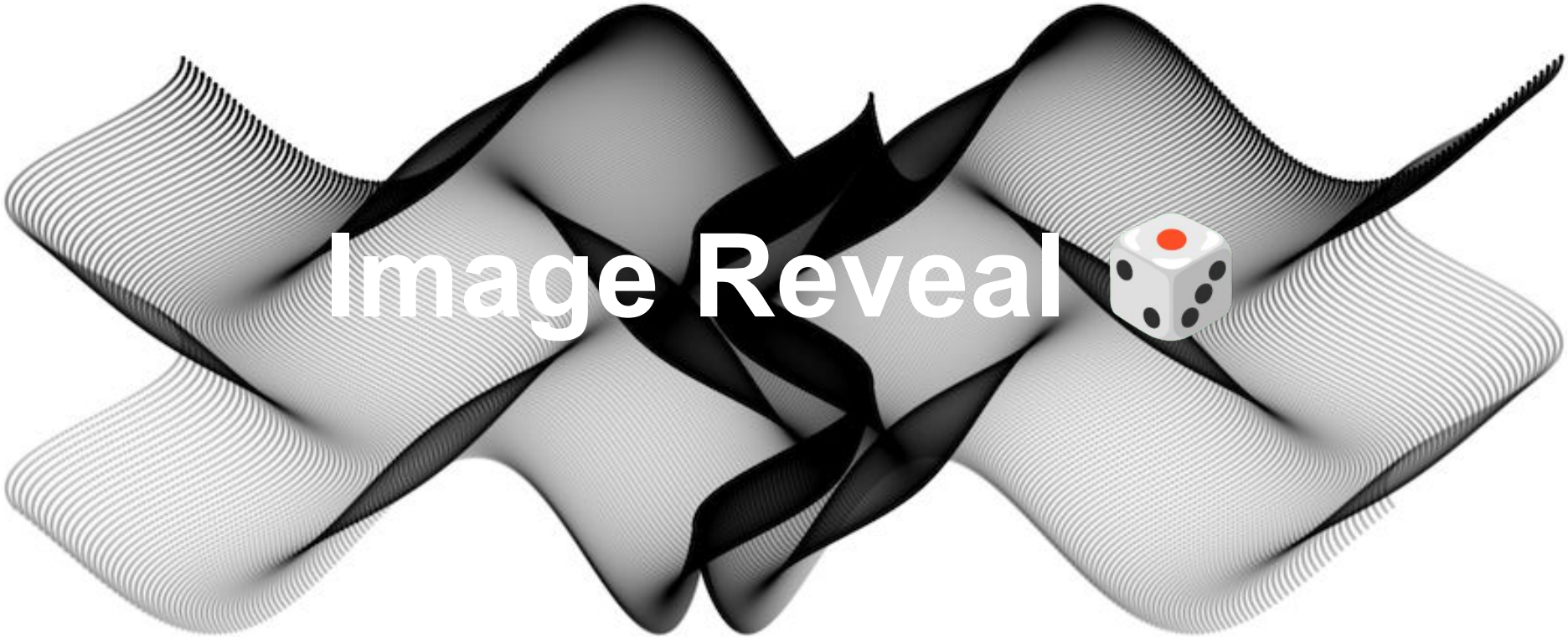
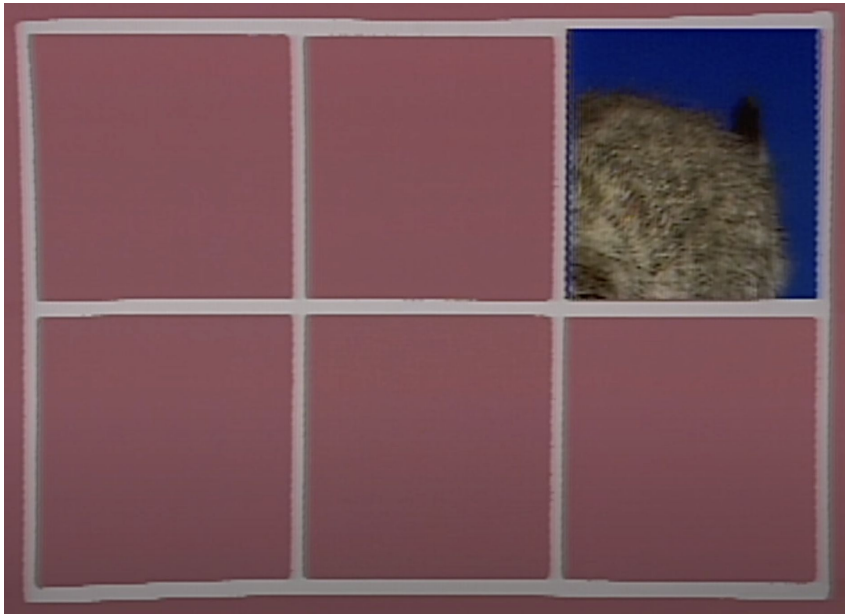

Image Reveal

Photo reveal



— Tik Tak

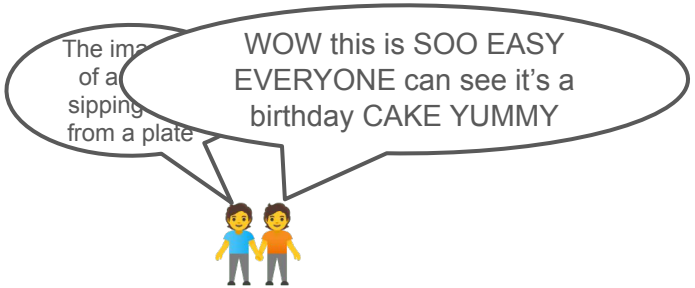
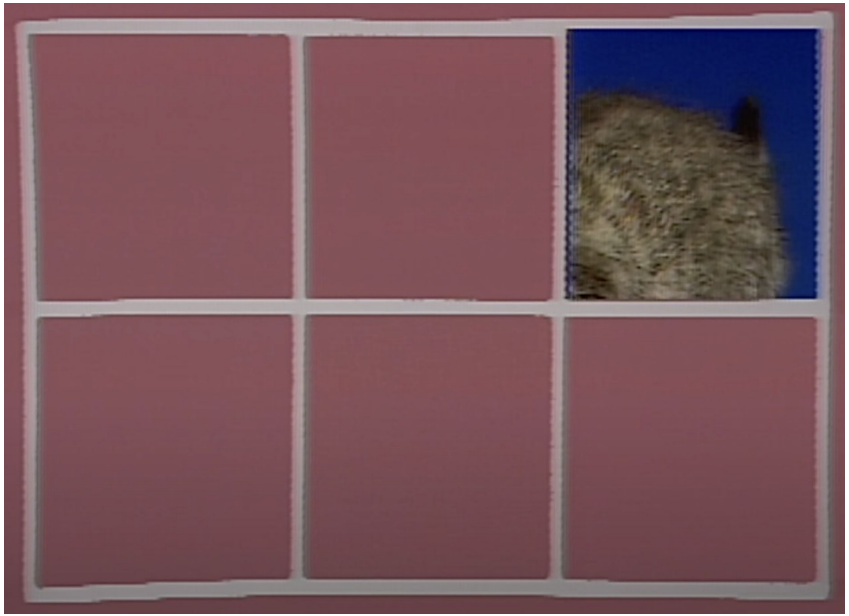


Photo reveal & statistical power

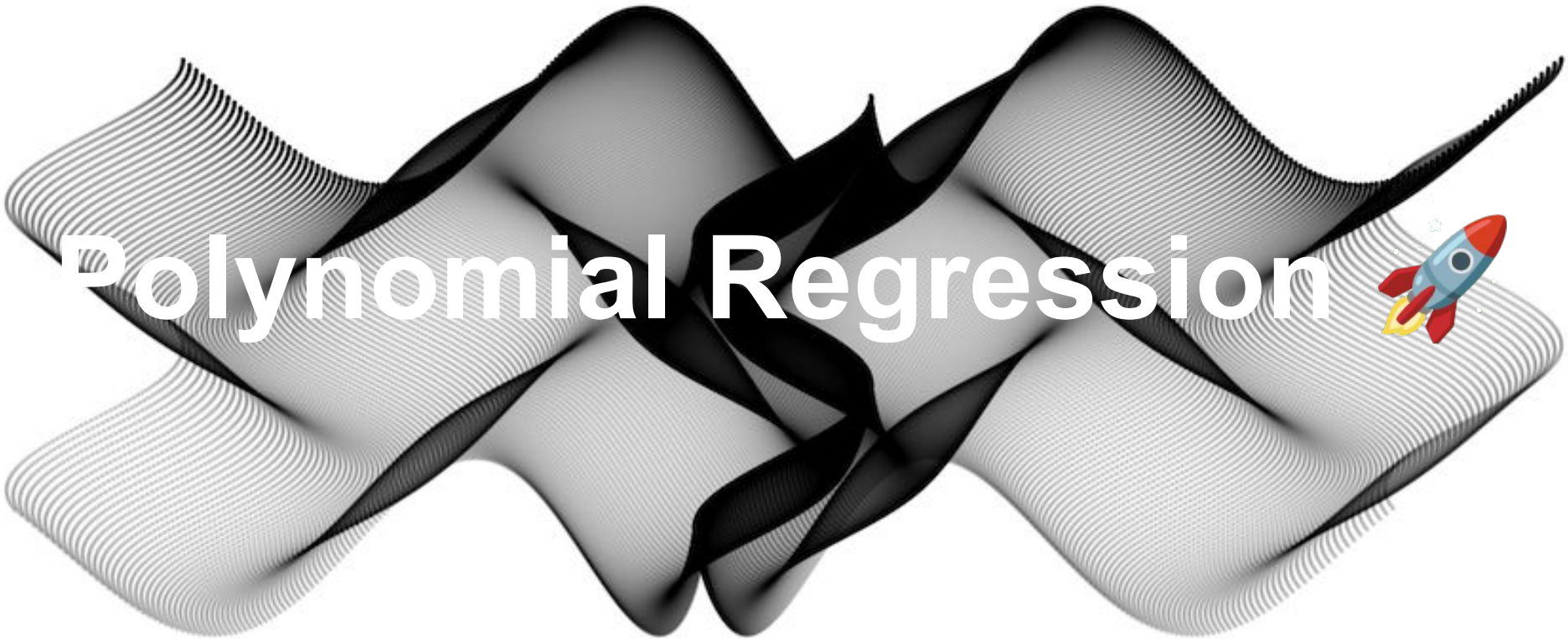


Number of observations

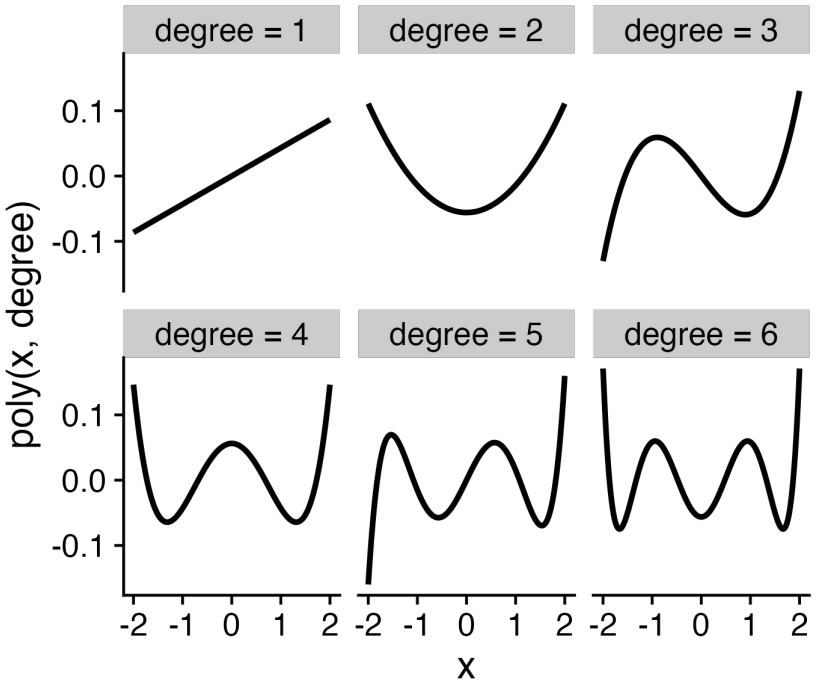


Noise level


Created with R


Polynomial Regression

Functional form | Nonlinear relationships

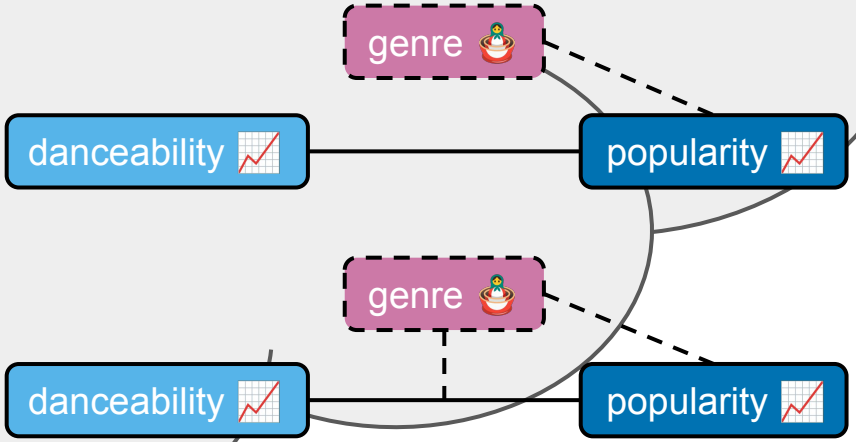


Danceability

"Danceability describes how suitable a track is for dancing based on a combination of musical elements including tempo, rhythm stability, beat strength, and overall regularity. A value of 0.0 is least danceable and 1.0 is most danceable."

— Spotify

- Does a hierarchical model make sense?
- What do we expect to be dependent?
  - Intercept: danceability depends on genre
  - Slope: the relationship depends on genre
  - Both



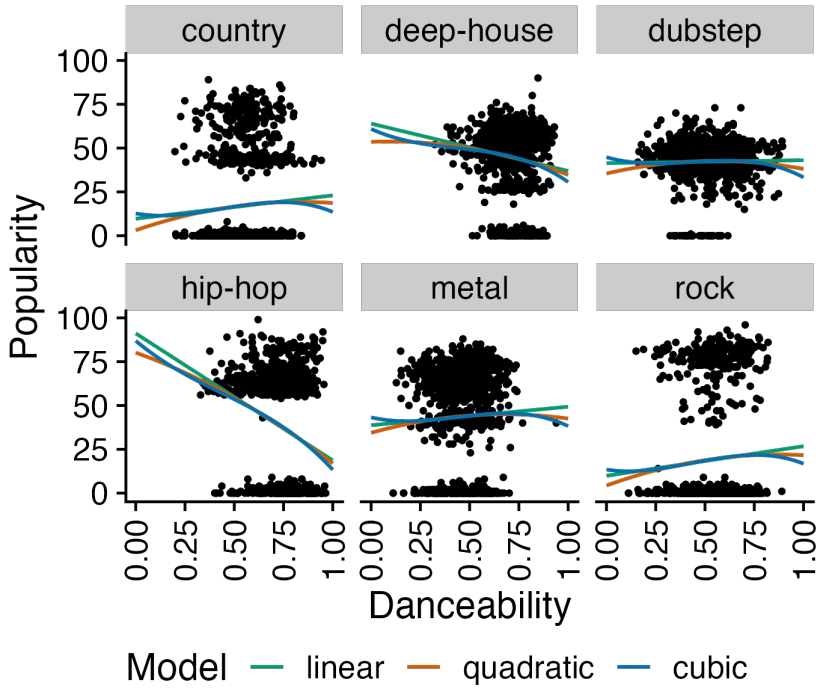
How are the danceability and popularity of tracks related?

Polynomial regression

```
mod_linear <- popularity ~ poly(danceability, 1) +
(1 + danceability | track_genre)
mod_quadratic <- popularity ~ poly(danceability, 2)
+ (1 + danceability | track_genre)
mod_cubic <- popularity ~ poly(danceability, 3) +
(1 + danceability | track_genre)

anova(fit_linear, fit_quadratic, fit_cubic)
```

```
Data: spotify_by_genre
Models:
fit_linear: mod_linear
fit_quadratic: mod_quadratic
fit_cubic: mod_cubic
              npar   AIC   BIC logLik -2*log(L)  Chisq Df Pr(>Chisq)
fit_linear       6 56362 56403 -28175      56350
fit_quadratic    7 56362 56409 -28174      56348 2.1599  1     0.1417
fit_cubic        8 56363 56416 -28174      56347 1.4380  1     0.2305
```

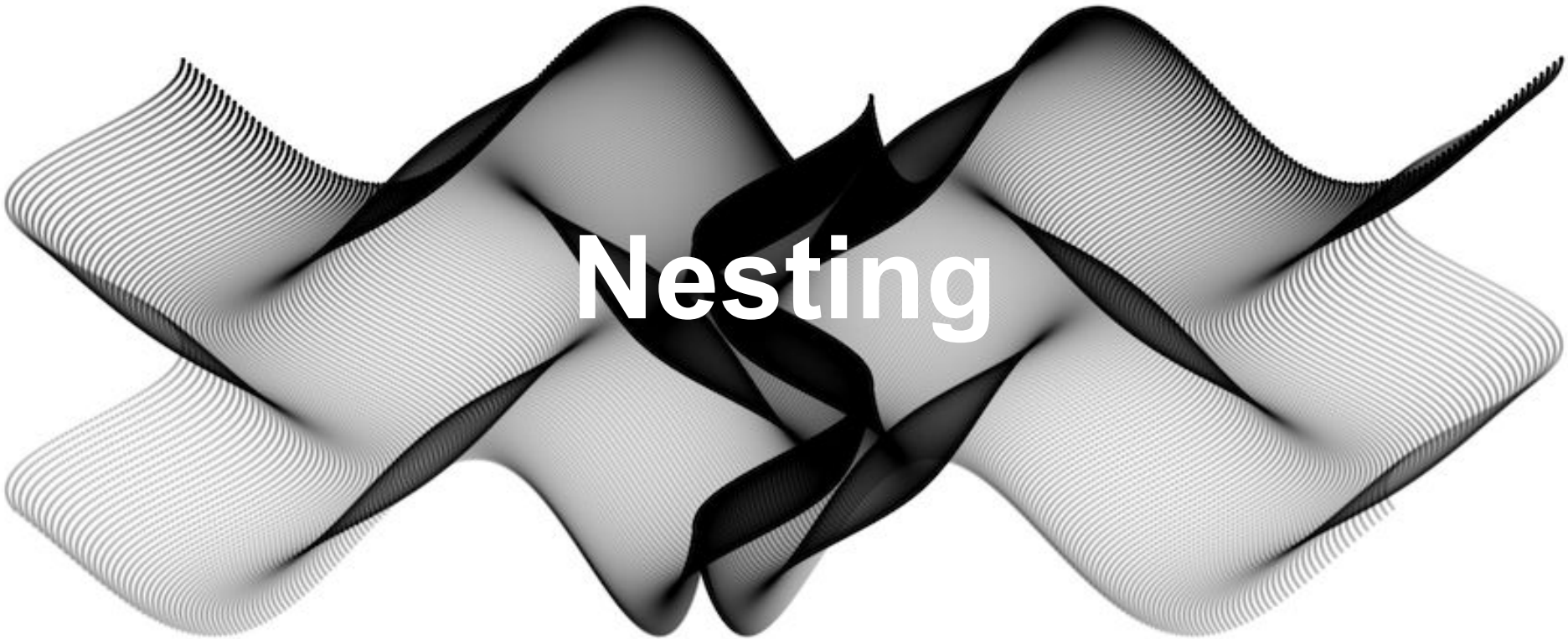


(Non-)linearity is not our biggest problem...

Quote of the week

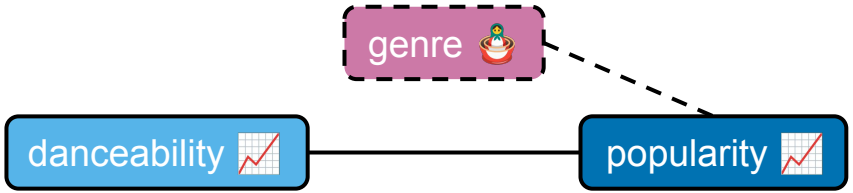
"All models are wrong, but some are useful."

– George Box (1976)

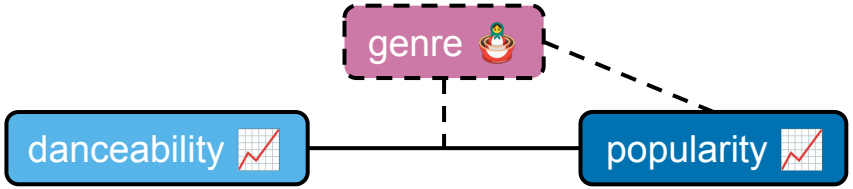

Nesting

Nested data

Nested intercept



Nested intercept + slope



Nested models

Nested models can be compared

- ❏ Are the parameters of the one model fully contained in the other?
- ❏ Use `insight::is_nested_models()` (only for multiple regression, ignores random effects parameters)

Nested 🪺 or unnested 🪺?

1. 🐣
   ```
   y ~ x1
   y ~ x1 + x2
   ```
2. 🍳
   ```
   y ~ x1
   y ~ x2
   ```
3. 🐣
   ```
   y ~ x1
   y ~ x2 + x1
   ```
4. 🐣
   ```
   y ~ x1
   y ~ x1 + x2 + x1:x2
   ```
5. 🍳
   ```
   y ~ x1 + x2 + x3
   y ~ x1 + x2 + x4 + x5
   ```
6. 🐣
   ```
   y ~ 1
   y ~ x1
   ```
7. 🍳
   ```
   y ~ 1 + x1
   y ~ 0 + x1 + x2
   ```

⚠️ The `anova` function doesn't complain if models are not nested.



Assumptions

regression diagnostics

“Conducting data analysis is like drinking a fine wine. It is important to swirl and sniff the wine, to unpack the complex bouquet and to appreciate the experience. Gulping the wine doesn’t work.”
— Daniel B. Wright (2003)

Validity

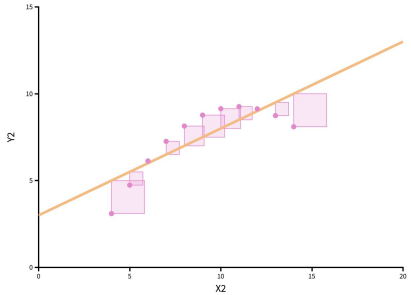
- Outcome reflects phenomenon of interest?
- Inputs are relevant and necessary?
- Sample represents population of interest?

⚠️ Important

```
predict(fit) # external validity
```

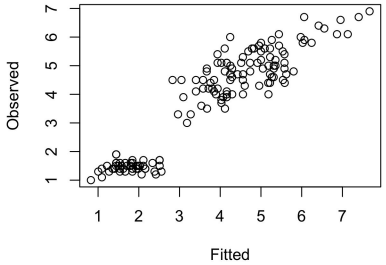
💡 Prioritization taken from Andrew Gelman.

Additivity & linearity

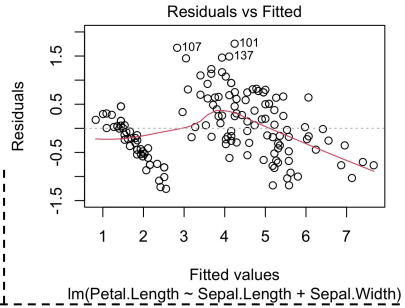


👍 Approximately linear; horizontal at 0
⚠️ Important

```
plot(x = fitted(fit), y =
iris$Petal.Length)
```




```
plot(x = fit, which = 1)
```




```
car::residualPlots(
model = fit) # for each
predictor
```

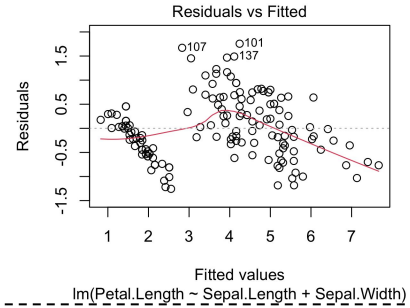
Independence of errors

✅ Independent observations
❌ Dependent observations

We know how to deal with this, do we? 🦸

👍 Horizontal
⚠️ Important

```
plot(x = fit, which = 1)
```

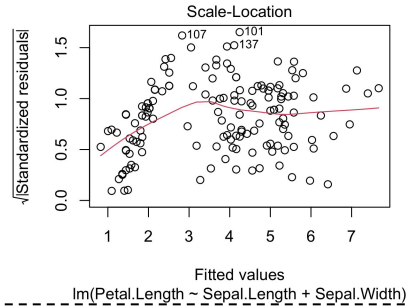


Equal variance of errors

Homogeneity of variance, homoscedasticity. Similar to sphericity in repeated measures ANOVA.

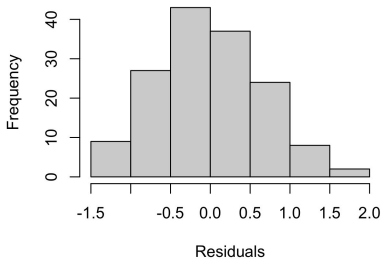
👍 Horizontal
⚠️ Issue with prediction, otherwise minor

```
plot(x = fit, which = 3)
```

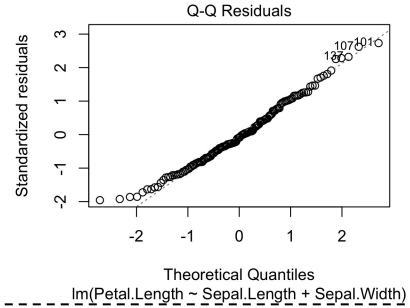


Normality of errors

```
hist(x = resid(fit))
```




```
plot(x = fit, which = 2)
```



👍 Approximately normal; linear

⚠️ Issue with predicting individual data points, otherwise not an issue

(Multi)collinearity

```
cor(iris[, c("Sepal.Length",
"Sepal.Width")])
```

```
car::vif(mod = fit)
```

👍 Low correlations between predictors; low VIF
⚠️ For explanation, less/not for prediction

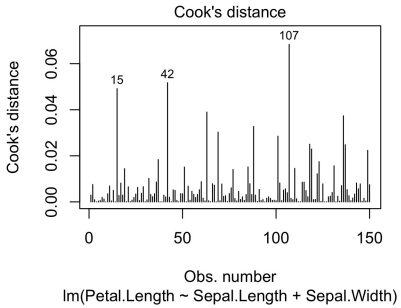
Influential observations

Influential observation = Outlier + Leverage

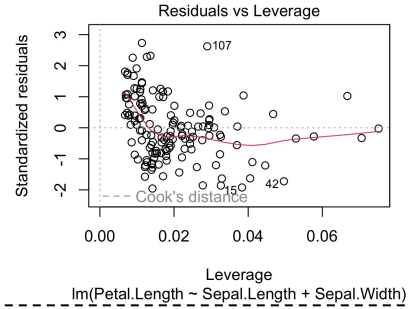
❌ Error
✅ Interesting
✅ Random

👍 Cook's distance < 1 or < 4/N; horizontal at 0
⚠️ It depends

```
plot(x = fit, which = 4)
```




```
plot(x = fit, which = 5)
```

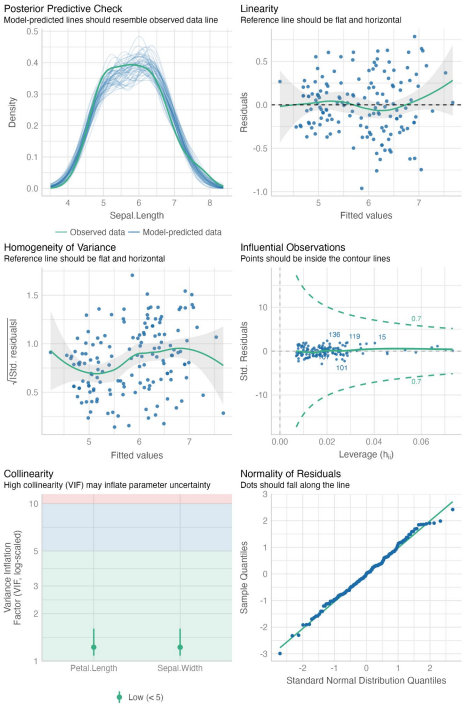


📖 Leys et al. (2019)

Take it easy

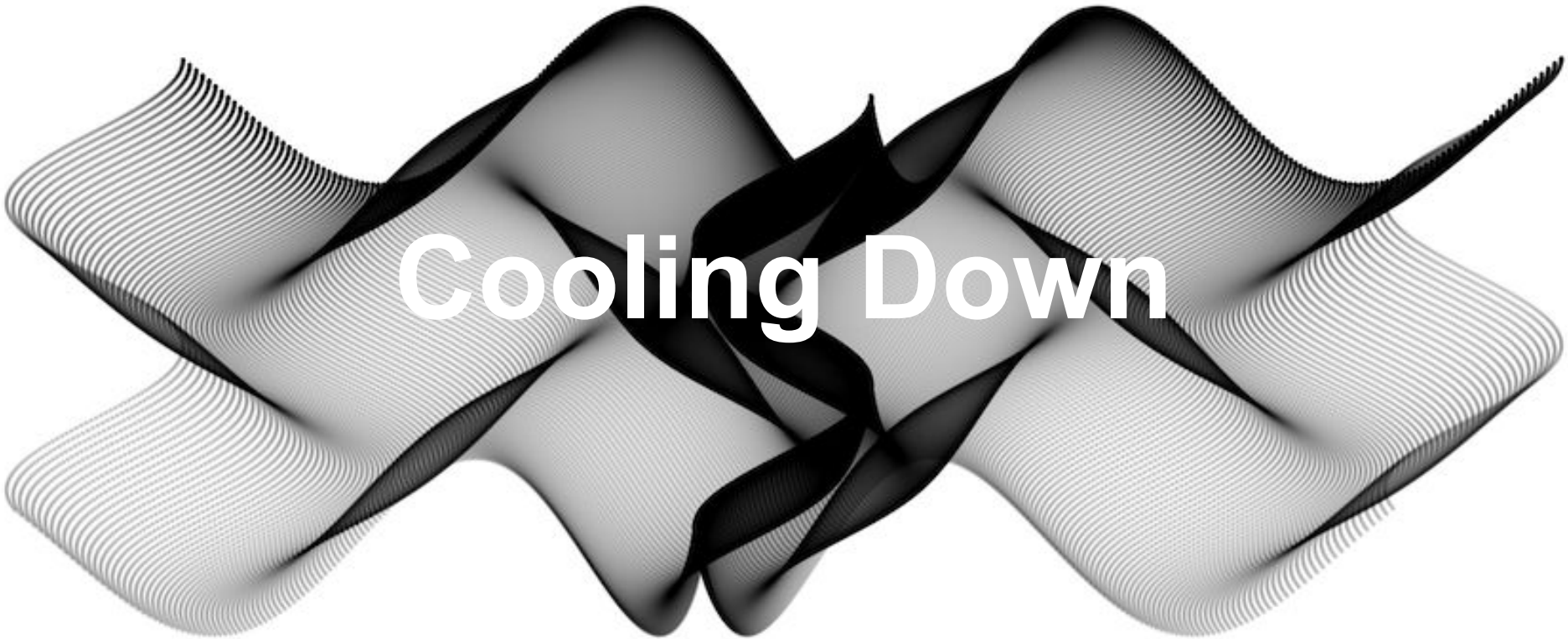
```
library("easystats")
performance::check_model(fit)

astatur::regression.diagnostics
```



[Interpretations and solutions]

📖 8.1.7 & 8.2


Cooling Down

Takeaways



Illustration by Amii Illustrates

Takeaways

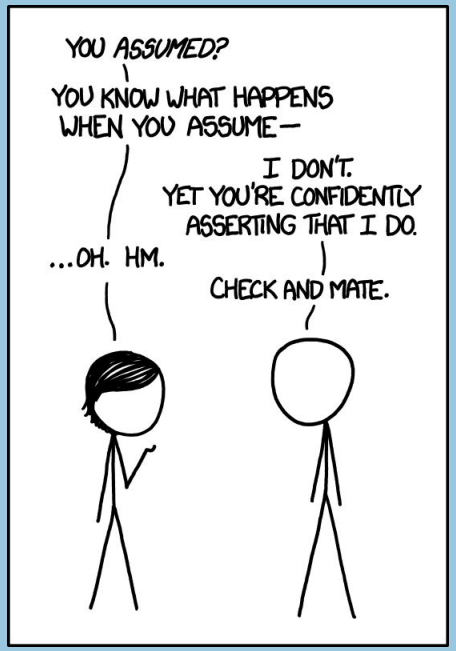


Illustration by Randall Munroe (wtf)

Takeaways

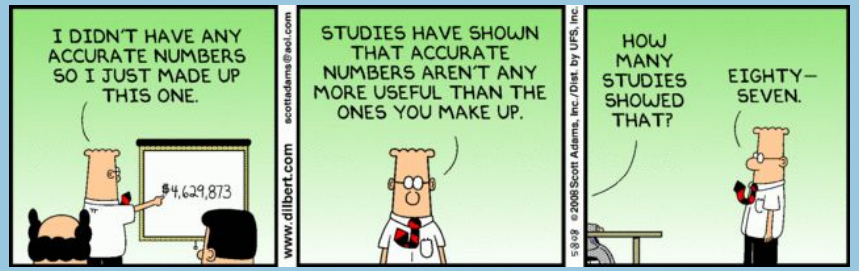


Illustration by Scott Adams

✍️ Colophon

Slides
alexandersavi.nl/teaching/

License
Statistical Reasoning by Alexander Savi is licensed under a Creative Commons Attribution-ShareAlike 4.0 International License. An Open Educational Resource. Approved for Free Cultural Works.